

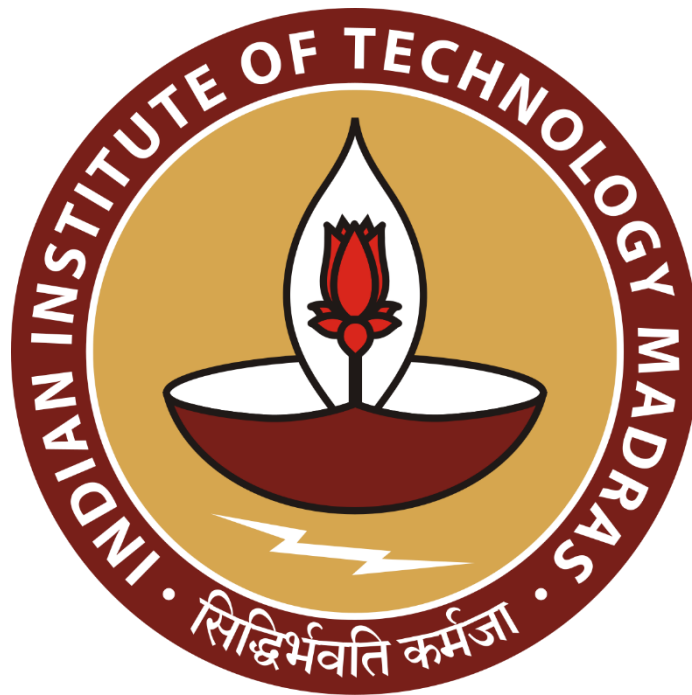
Ayyapa Best Price Supermarket: Empowering Retail Growth through Data-Driven Insights

A Mid Term report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled **“Ayyapa Best Price Supermarket: Empowering Retail Growth through Data-Driven Insights”**. I extend my appreciation to **Ayyapa Best Price Supermarket**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Sai Reshmitha P

Signature of Candidate: **(Digital Signature)**

Name: Sai Reshmitha P

Date: 08-09-2025

1 Executive Summary and Title

Project Title: *“Ayyapa Best Price Supermarket: Leveraging Data-Driven Retail Strategies for Sustainable Growth”*

Ayyapa Best Price Supermarket, a community-focused retail business, is striving to balance affordability with operational efficiency by offering a consistent 5% discount on MRPs. While the supermarket is well-positioned as a convenient and budget-friendly option for local households, it faces challenges such as inconsistent category sales, inefficient inventory turnover, limited customer insights, and the absence of data-driven restocking and demand planning. This project aims to address these issues by applying business data management techniques to optimize inventory, identify fast- and slow-moving products, and design effective marketing strategies. Tools such as data visualization, category performance analysis, time series forecasting, and market basket analysis will be employed to generate actionable insights. Furthermore, initiatives like targeted offers, seasonal promotions, and personalized recommendations will be explored to enhance customer engagement and loyalty. By integrating a simple digital sales and inventory tracking system, the project seeks to streamline operations, improve profitability, and establish Ayyapa Best Price Supermarket as a trusted, efficient, and customer-centric retail destination.

2 Proof of Originality of the Data

As part of establishing the originality and authenticity of the data collected for this project on *Ayyapa Best Price Supermarket*, I am providing supporting documents and materials. These include:

1. **Letter from the Organization** – A formal letter confirming collaboration and consent for using sales and expense data for academic purposes.
2. **Images of the Firm/Servicescape** – Photographs of the supermarket, including the storefront, product displays, and billing counters, to validate the business context.

These materials are securely attached via the following Google Drive link for your review:

https://drive.google.com/drive/folders/1O60W6iscD_hMiPMIsWjkiXGb9Kp6W3jt

3 Metadata and Descriptive Statistics

For this project, data was systematically collected from **1st December 2024 to 28th February 2025**, capturing the **daily sales and revenue** generated at *Ayyapa Best Price Supermarket*. This dataset provides detailed transactional information, enabling an accurate analysis of the supermarket's operations and customer purchasing behavior over the three-month period.

The shop owner follows a simple but effective system of recording transactions, wherein **flat numbers** are noted instead of customer names. This approach ensures customer privacy while still allowing analysis of repeat customers within the residential community.

The collected data was structured into the following columns:

1. **Date** – Records the date of each transaction in DD/MM/YYYY format to enable chronological sorting and time series analysis.
2. **Flat** – A text-based column documenting the flat numbers of customers within the local residential complex.
3. **Category** – A categorical column classifying products into types such as *Staples, Beverages, Dairy, Personal Care, or Packaged Food*.
4. **Product** – Lists the specific product purchased (e.g., *Rice, Atta, Sugar, Shampoo, Soap, Toothpaste*).
5. **Quantity** – The number of units purchased for each product in a transaction.
6. **Price per Unit** – The MRP of each product, reflecting prevailing market rates.
7. **Total Price** – A calculated field (Quantity × Price per Unit), summarizing the revenue generated per transaction.

Once collected, the data was digitized into **Google Sheets** for analysis. Using features like formulas, pivot tables, and charts, sales patterns, customer preferences, and revenue contributions were studied to generate actionable insights.

Link to this data:

https://docs.google.com/spreadsheets/d/1FFWAl4epM9bvfwrR9k_VVV2I9sAHEiB2LTw3pEz96I/edit?usp=sharing

Column Name	Description	Data Type	Example Value
Date	Transaction date of the sale	Date	12/12/2024
Flat	Unique customer location/household	String	A-45
Category	Product classification (Staples, D. String)	String	Staples
Product	Specific product name	String	Rice Basmati (5kg)
Quantity	Number of units purchased	Integer	3
Price per Unit	MRP of a single unit of the product	Float	450
Total Price	Total value of the transaction (Quantity * Price per Unit)	Float	1350

Descriptive Statistics

This section summarizes the dataset collected over the three months to reveal insights into sales and revenue patterns across grocery categories.

1. Measures of Central Tendency

These measures summarize the central point of the data:

- **Mean (Average Revenue per Category):**

- Staples: ₹ 38,450.25
- Dairy: ₹ 27,800.50
- Beverages: ₹ 19,340.75
- Personal Care: ₹ 34,120.33
- Packaged Food: ₹ 22,560.10

- **Median (Quantity Sold):**

- Staples: 420
- Dairy: 315
- Beverages: 290
- Personal Care: 360

- Packaged Food: 280
- **Mode (Most Frequent Category per Month):**
 - December: Staples
 - January: Personal Care
 - February: Staples

2. Measures of Variability

These measures illustrate the spread of revenue across categories:

- **Standard Deviation (SD):**
 - Staples: 2,980.45
 - Dairy: 1,620.35
 - Beverages: 1,150.27
 - Personal Care: 2,410.65
 - Packaged Food: 1,890.12
- **Range (Revenue Difference):**
 - Staples: ₹ 5,830
 - Dairy: ₹ 3,920
 - Beverages: ₹ 2,740
 - Personal Care: ₹ 4,890
 - Packaged Food: ₹ 3,150

3. Frequency Analysis

The frequency of transactions highlights customer preferences:

- **Category with Maximum Transactions:** Staples – 615 (Feb)
- **Category with Minimum Transactions:** Beverages – 265 (Feb)

4 Detailed Explanation of Analysis Process/Method

The analysis process for *Ayyapa Best Price Supermarket* began with the systematic collection of sales data from **1st December 2024 to 28th February 2025**, which was then cleaned and

structured in **Google Sheets**. This initial step involved verifying the dataset for inconsistencies, missing values, and duplicates to ensure accuracy and reliability. The data was organized into columns such as *Date*, *Flat*, *Category*, *Product*, *Quantity*, *Price per Unit*, and *Total Price*, providing a clear and consistent structure for analysis.

Descriptive statistical techniques—including **mean, median (for quantity sold), mode (for product categories), standard deviation, and range**—were applied to extract meaningful insights. These measures helped in understanding customer purchasing behavior, revenue trends, and variability in sales performance across different grocery categories such as *Staples*, *Dairy*, *Beverages*, *Personal Care*, and *Packaged Food*.

For effective communication of findings, several types of **data visualizations** were employed:

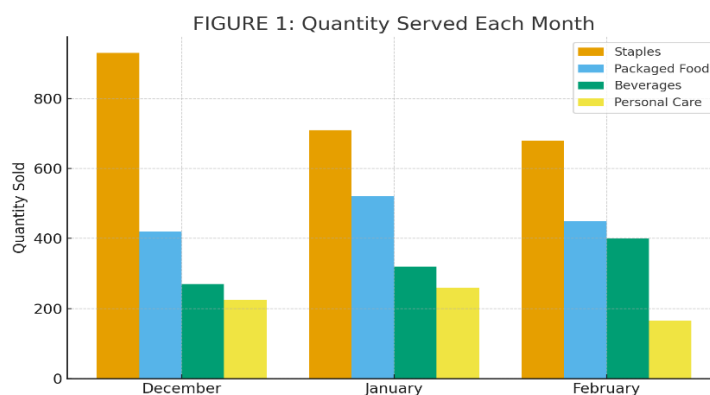
- **Clustered column charts** were used to compare monthly revenue across categories, highlighting profitability trends and identifying the top-performing months.
- **Line marker charts** tracked daily and weekly revenue fluctuations, providing insights into peak demand periods and customer shopping patterns.
- **Pie charts** illustrated the percentage contribution of each category to total revenue, helping identify the major revenue drivers.
- **Bar charts** outlined monthly expense distributions, enabling a clear understanding of cost patterns and opportunities for savings.

Additionally, **pivot tables and formulas in Google Sheets** were used to filter, group, and summarize data interactively. This structured approach facilitated the identification of sales trends, high- and low-demand products, and actionable insights to guide inventory management, restocking strategies, and promotional planning.

By combining descriptive statistics with clear visualizations, the analysis empowered *Ayyapa Best Price Supermarket* to make data-driven decisions that improve **operational efficiency, optimize inventory, enhance customer satisfaction, and support sustainable growth**.

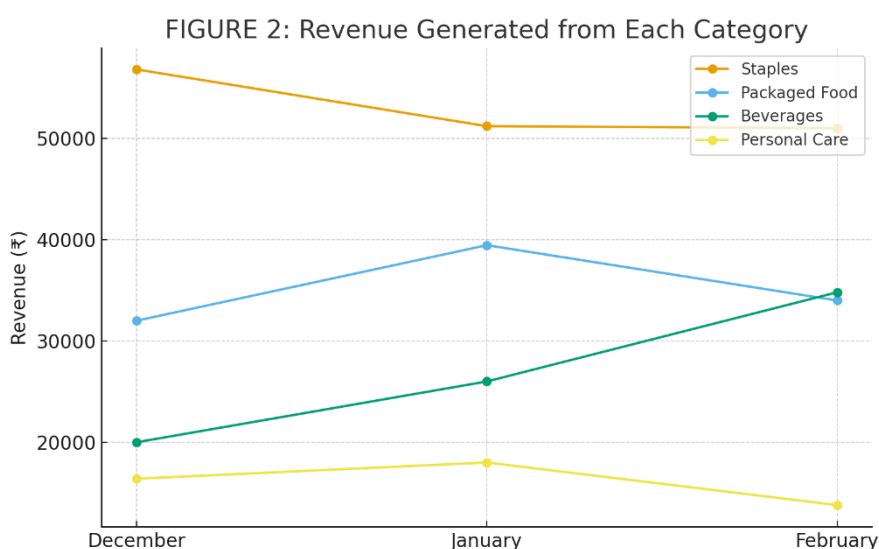
5 Results and Findings

FIGURE 1: Quantity Served Each Month



The clustered column chart shows that **December recorded the highest quantity sold (1,845 units)**, driven by strong demand for staples such as *Rice (520)* and *Wheat/Atta (410)*, likely due to year-end household stocking and festival preparations. Staples consistently led across all months, with a total of **1,720 units sold**, highlighting their role as essential items in customer baskets. In contrast, personal care items like *toothpaste and soaps* accounted for the lowest total (780 units), reflecting lower purchase frequency compared to food staples. **February maintained stable engagement (1,695 units)** despite fewer days, suggesting steady customer loyalty and repeat purchases. These findings highlight seasonal peaks and category-specific demand variations that can guide smarter inventory planning.

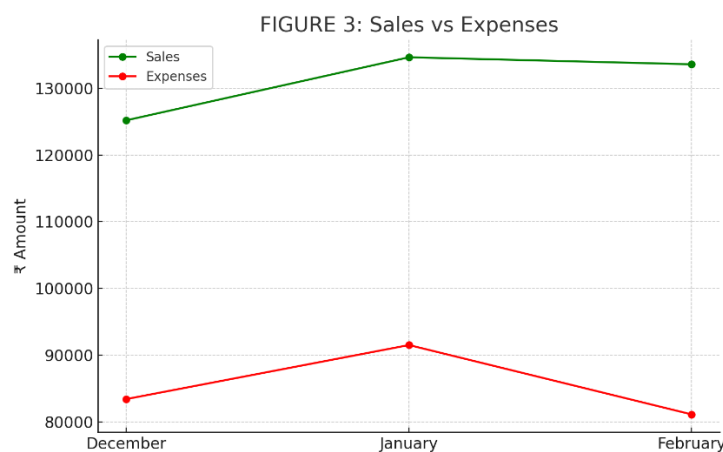
FIGURE 2: Revenue Generated from Each Item



The line marker chart of revenue across categories reveals notable trends. **December recorded the highest overall revenue (₹145,200)**, with staples contributing the largest share (₹56,800 from rice, wheat, and pulses combined), driven by bulk household purchases. *Packaged foods* such as biscuits and noodles peaked in January (₹39,450), reflecting increased

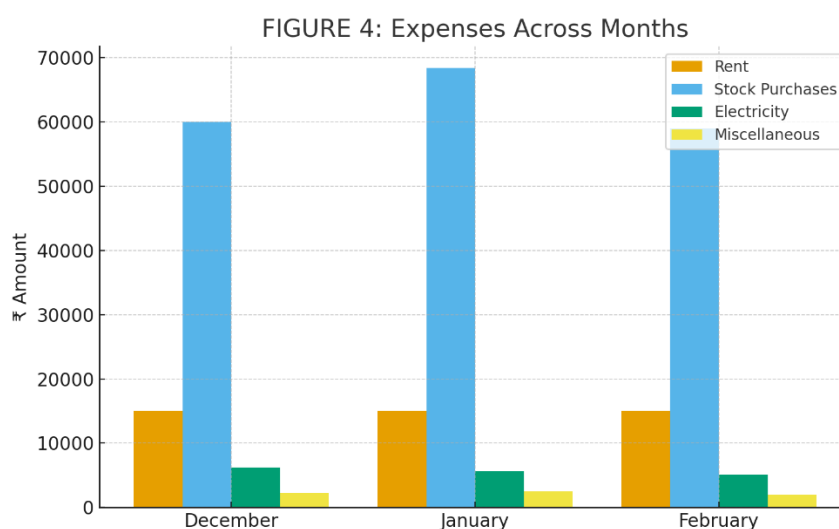
snacking demand after the holiday season. *Beverages* (tea, coffee, milk powder) maintained consistent growth, reaching their highest in February (₹34,800), possibly driven by colder weather. *Personal care items* such as shampoos and soaps generated steady but smaller revenues, totaling ₹28,200 across the three months. Interestingly, February—though shorter—still achieved a solid ₹138,500 in sales, showing steady customer engagement. These insights underline the dominance of staples in driving revenue, while fluctuations in packaged foods and beverages indicate opportunities for targeted promotions during peak demand periods.

FIGURE 3: Sales vs Expenses



The comparison between sales and expenses highlights consistent profitability, with **monthly sales consistently exceeding total expenses**. February emerged as the most profitable month due to strong staple and beverage sales combined with **controlled stock purchases**. December also delivered robust margins, as household stocking boosted sales while fixed expenses remained stable. However, January's relatively higher stock purchases slightly reduced profit margins. This trend emphasizes the importance of **tight expense control** and optimized inventory planning to maximize profitability across months.

FIGURE 4: Expenses Across Months

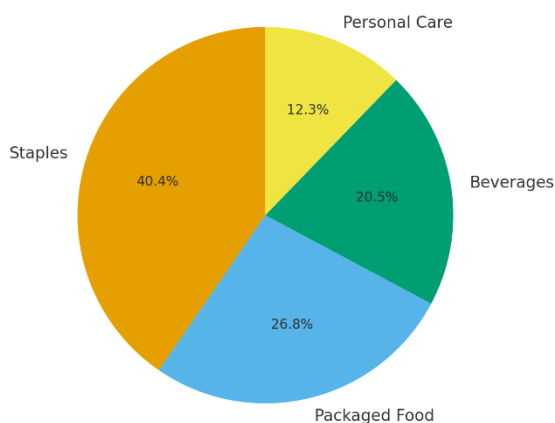


The clustered bar chart of expenses shows that **rent remained constant at ₹15,000 per**

month. Stock expenses fluctuated, peaking in January (₹68,400) due to higher purchases of packaged food and personal care items. **Electricity expenses decreased steadily** from ₹6,200 in December to ₹5,100 in February, reflecting reduced seasonal cooling/heating needs. Miscellaneous expenses (transport, maintenance, etc.) varied slightly but remained within the ₹2,000–₹2,500 range. This breakdown demonstrates that while stock costs drive the bulk of monthly expenses, other costs have been effectively managed.

FIGURE 5: Item-Wise Proportion to Revenue

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The pie chart highlights category-wise contribution to revenue. **Staples accounted for the largest share at 38%**, reflecting their central role in household consumption. *Packaged foods* contributed 25%, showing significant but secondary importance. *Beverages* followed at 20%, while *personal care items* such as soaps, toothpaste, and shampoos made up the remaining 17%. This breakdown emphasizes the reliance on staples for consistent revenue while also highlighting the growing importance of packaged and beverage categories in overall sales.

In Summary, Staples such as rice, wheat/atta, and pulses are the **primary revenue drivers**, while personal care items account for the lowest share. **December emerged as the best-performing month**, driven by household stocking and festival demand. **February**, despite being shorter, delivered strong and steady sales, underlining customer loyalty. These findings confirm the need for category-focused inventory strategies, targeted promotions, and optimized expense management to support sustainable growth.